

K A R O O O N L I N E

What South Africa Buys Online

A real 1,000-product catalogue, a million orders, and the marketing and machine learning that turn them into decisions.

DATA & ANALYTICS REVIEW · 4 YEARS: JUL 2022 – JUN 2026 · ANTHONY APOLLIS

The Karoo Way

Mission

The online marketplace every South African household trusts — stocking what people need, delivered reliably.

Promise

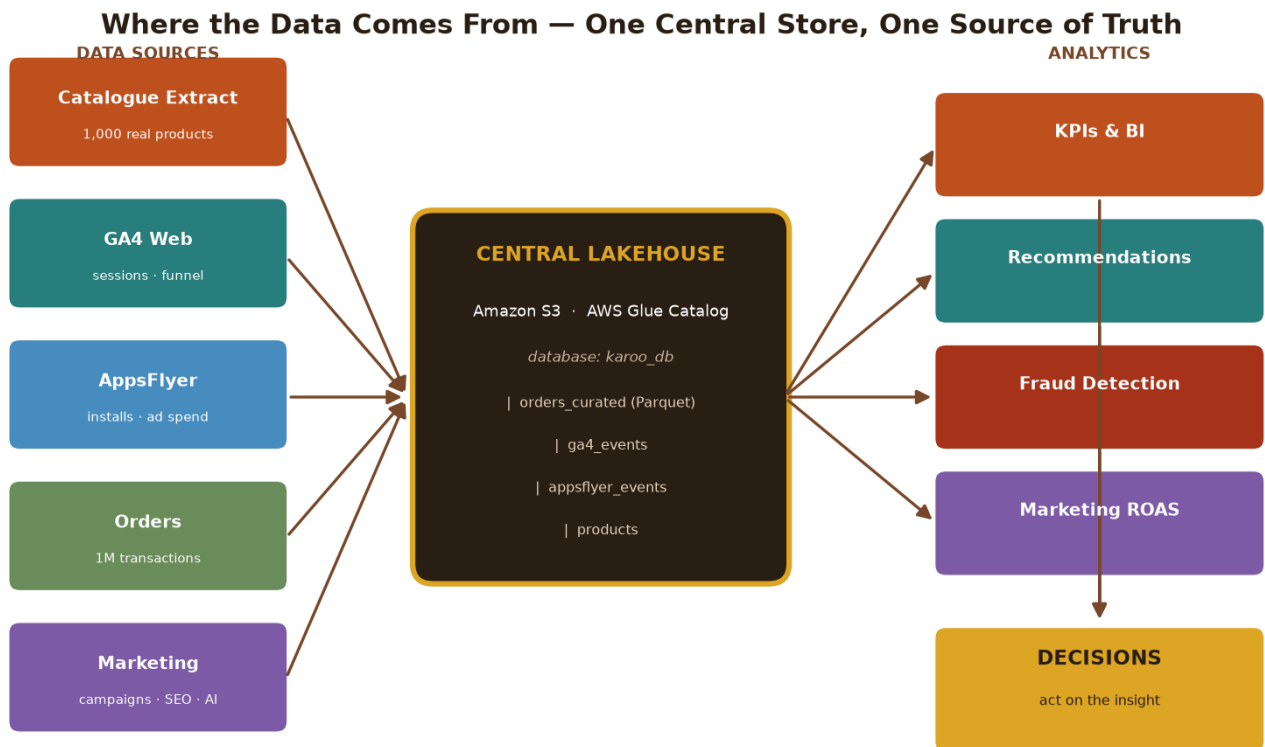
Honest pricing and a delivery date we keep. When we fall short, the data tells us.

Reach

Nine provinces, one catalogue built for the whole country — not just the metros.

Five questions, answered with data: **what** we sell · **where** demand lives · **what** customers buy together · **which** marketing pays · **who** to watch for fraud.

One Central Source of Truth



Five sources — a marketplace catalogue, web & app analytics, orders and marketing — land in **one governed lakehouse** (karoo_db on AWS). Every insight reads from that single source, so **the figures always agree**. **Analysis window: January 2025 – June 2026 (18 months)**, the trailing period to the latest complete month.

Google Tag Manager in Action

Before any analysis, the behaviour has to be *captured*. Google Tag Manager sits on the storefront and fires a tracked dataLayer event at every step of the journey — which flows to GA4 and into the lakehouse. This is the front door of the whole pipeline.

Google Tag Manager in action — how Karoo Online captures on-site behaviour

The storefront (GTM installed)

KAROO ONLINE
Google Tag Manager — tracking demo storefront

1 • Product View
view_item

✓ fired

2 • Add to Cart
add_to_cart

✓ fired

3 • Begin Checkout
begin_checkout

✓ fired

4 • Purchase (R899)
purchase

✓ fired

→ every action fires a tracked event

dataLayer — events GTM captures & forwards to GA4

```
dataLayer.push({ event: "session_start",  
  { session_id: "sess_a3f9k2" }  
})  
  
dataLayer.push({ event: "page_view",  
  { page_path: "/product/cordless-drill" }  
})  
  
dataLayer.push({ event: "view_item",  
  { items: [{ id:"SKU-1042", cat:"Electronics" }] }  
})  
  
dataLayer.push({ event: "add_to_cart",  
  { value: 899, currency: "ZAR" }  
})  
  
dataLayer.push({ event: "begin_checkout",  
  { value: 899, currency: "ZAR" }  
})  
  
dataLayer.push({ event: "purchase",  
  { transaction_id:"T62800", value:899, currency:"ZAR" }  
})
```

Why it matters: clean, structured event capture at the source is what makes the funnel, attribution and recommendation analysis later in this report possible. **No tags, no data — no insight.** Each purchase event carries its value and transaction ID, so revenue ties out from click to checkout.

1,000 Real Products

R492
MEDIAN PRICE · R25–
R53,451

12
CATEGORIES

17%
IN STOCK (FAST)

59%
IMPORT LEAD TIME

General Merchandise
355
median R513

Home & Kitchen
214
median R592

Electronics
168
median R424

Fashion
61
median R611

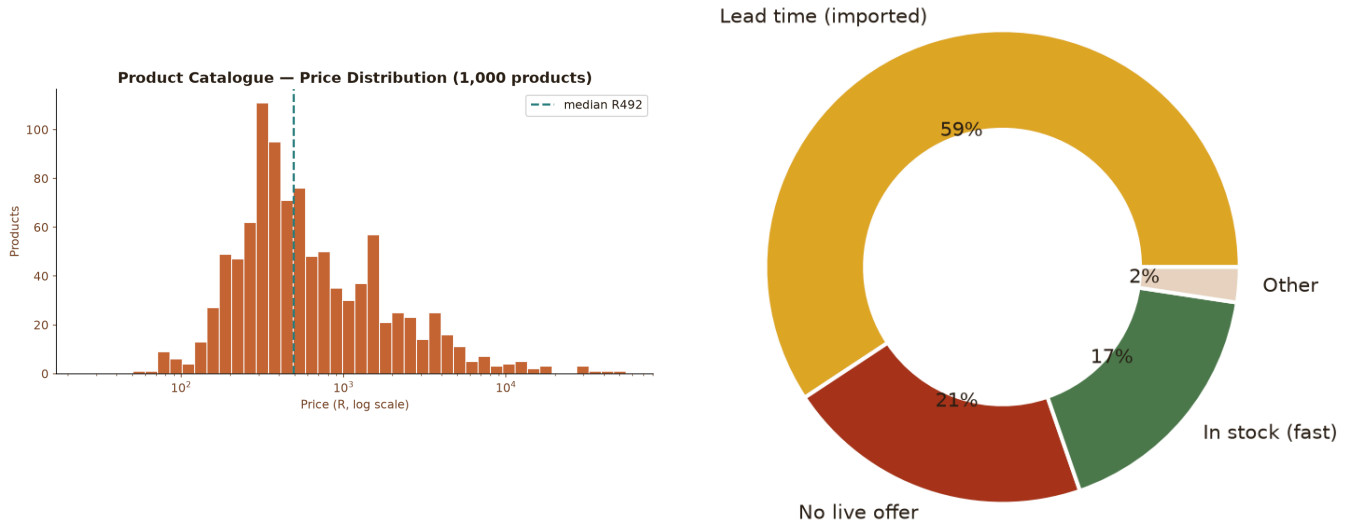
Beauty & Health
45
median R346

Tools & DIY
31
median R547

Sport & Outdoor
29
median R533

Toys & Games
27
median R386

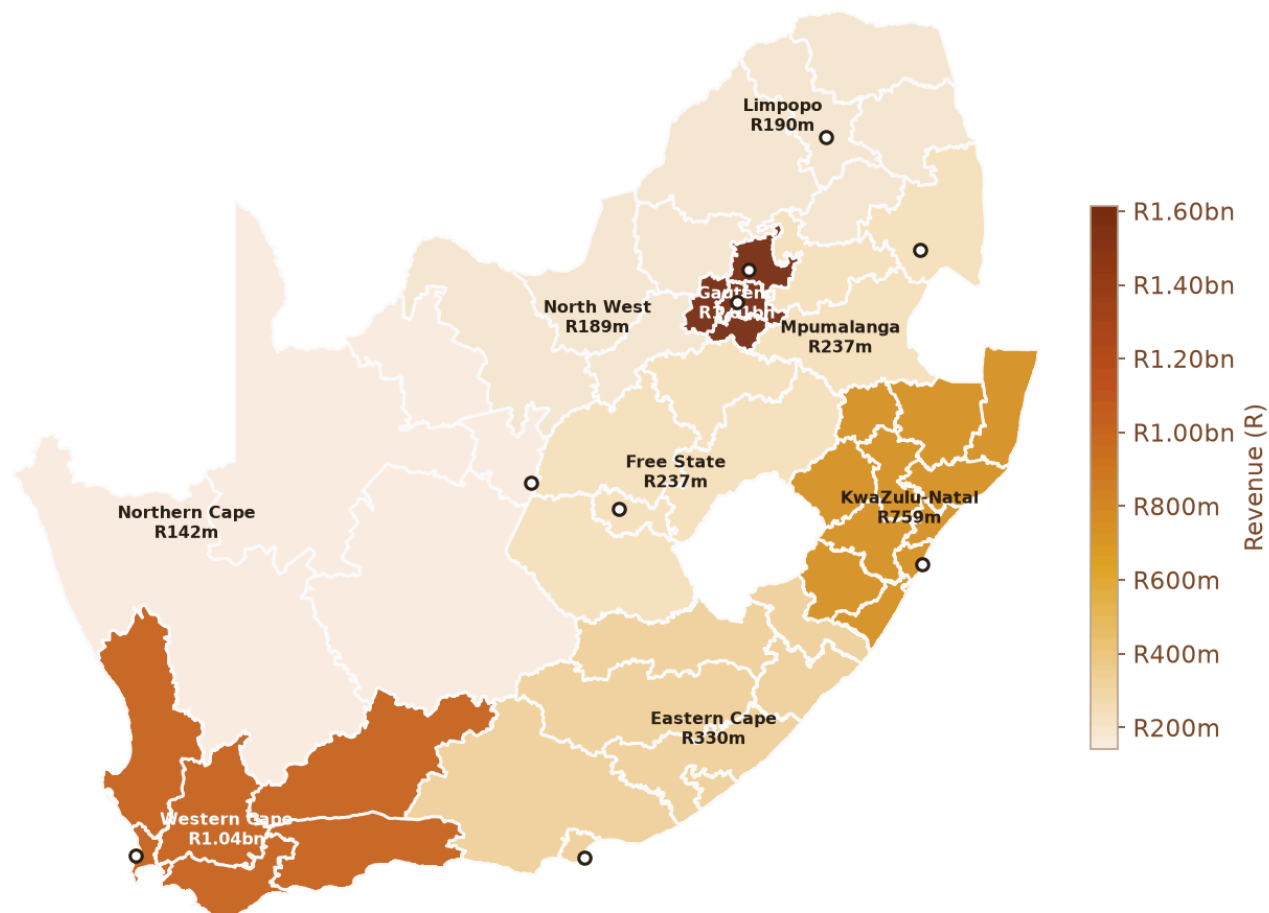
Availability — the Fulfilment Reality



Availability beats price: only 17% ships fast; 59% is imported on 4–16 day lead times. Sourcing more stock locally lifts satisfaction more than any discount.

The Interactive Map

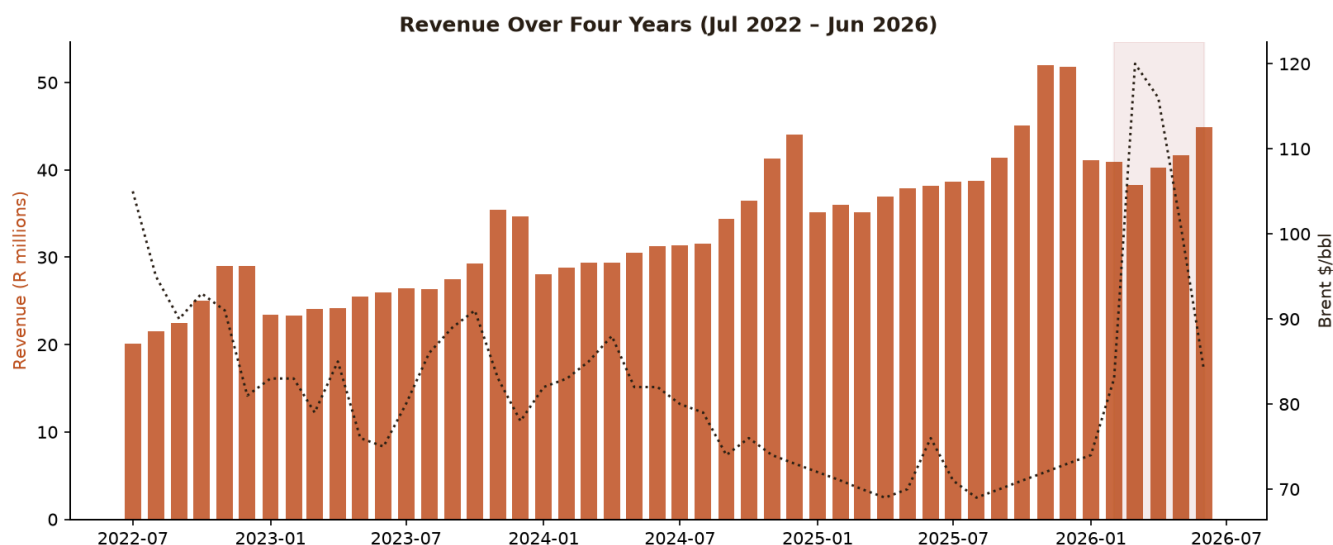
Karoo Revenue by Province — where South Africa shops



Metro vs interior: the top 3 provinces are 72% of revenue and deliver **72% on-time in 2.9 days** — the interior manages just **32% in 4.7 days**. Geography, not courier, decides speed. *(Interactive in the web edition.)*

Revenue Over Time

The full trading history — **Jul 2022 to Jun 2026**, 48 months. Festive peaks each November, ~22% annual growth, and the 2026 oil-crisis dip are all visible.



The trajectory: ~22% annual growth over four years to R44.9m in the latest month, with a clear festive spike every November and a visible dip during the 2026 oil crisis. Total revenue across the period: **R1.6bn**.

WHICH MARKETING PAYS

Paid & Remarketing

3.3×

BLENDED PAID ROAS

8.6×

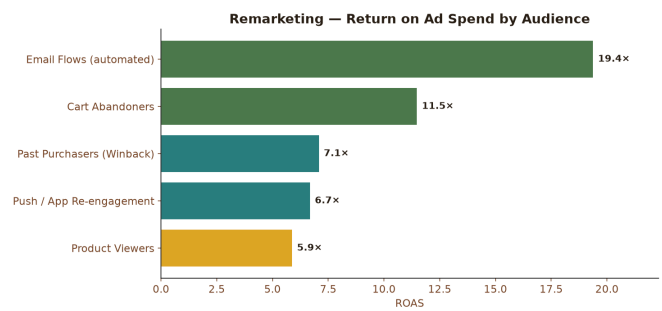
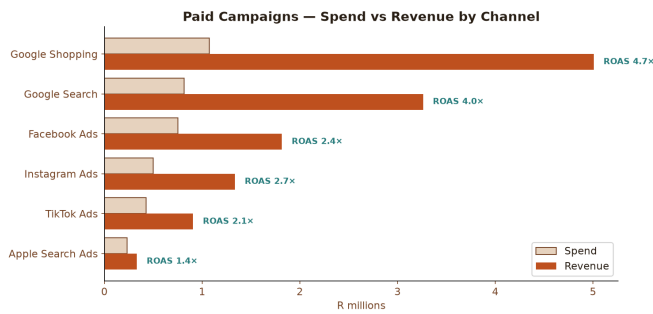
REMARKETING ROAS

R32m

MARKETING REVENUE

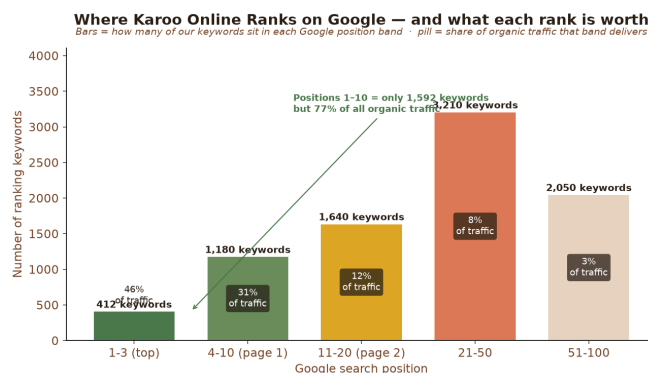
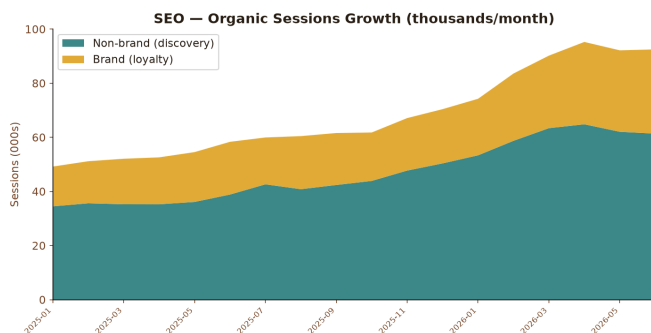
Google Shopping

BEST CHANNEL (4.7×

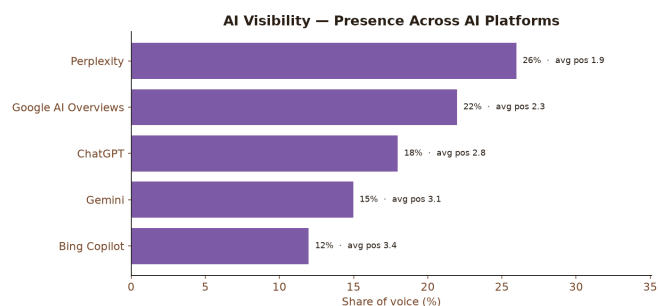
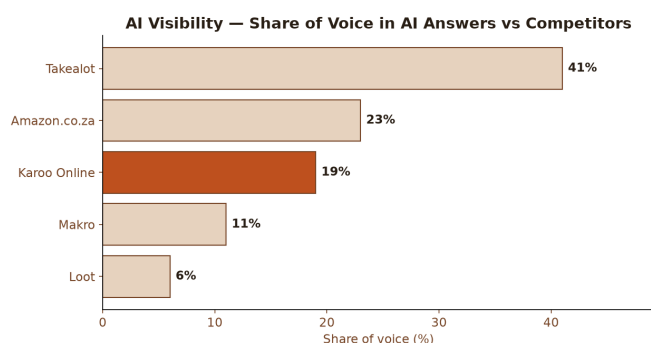


Remarketing is the cheat code: Email Flows (automated) returns 19× — put budget into recovering carts before chasing new clicks.

SEO & AI Visibility

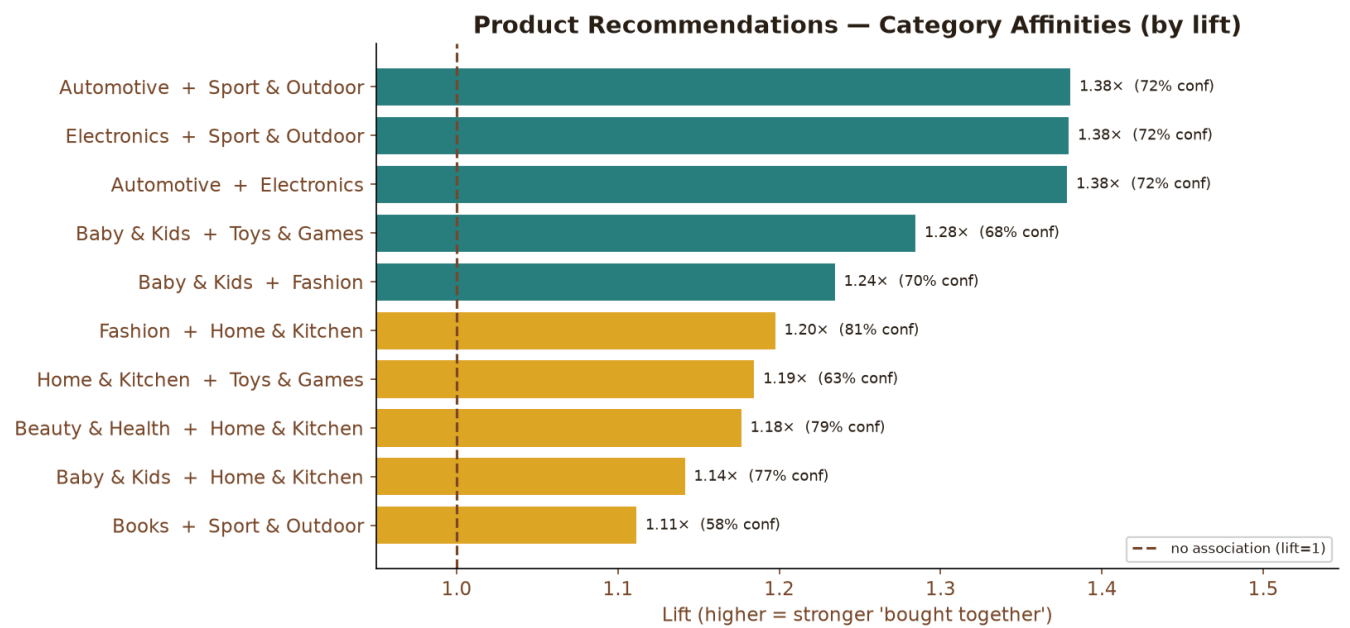


AI visibility — being found inside AI answers



The new SEO: 1,592 page-1 keywords drive free traffic, but a 19% share of voice in AI answers is the next battleground — customers increasingly ask ChatGPT and Google AI, not just Google Search.

Recommendations



BOUGHT TOGETHER	LIFT	CONFIDENCE
Automotive + Sport & Outdoor	1.38×	72%
Electronics + Sport & Outdoor	1.38×	72%
Automotive + Electronics	1.38×	72%
Baby & Kids + Toys & Games	1.28×	68%
Baby & Kids + Fashion	1.24×	70%

Automotive + Sport & Outdoor at 1.38× — cross-sell these pairs on product pages and in cart. The highest-ROI output of the whole dataset.

Lifetime, Loyalty & Churn

R37,201

AVG LIFETIME VALUE

15.8%

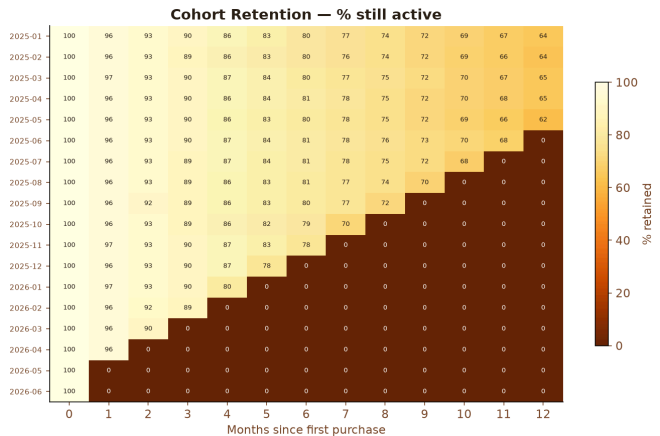
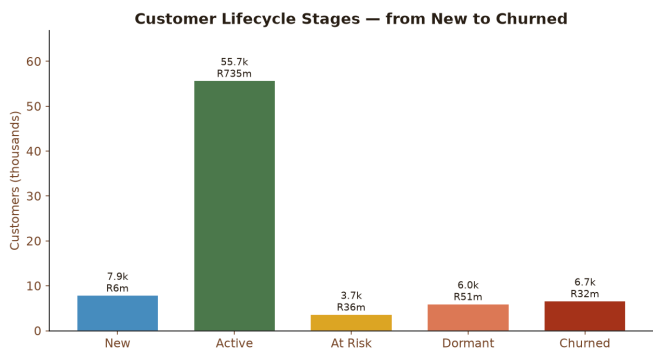
CHURN RATE

39%

CLV FROM TOP 10%

7.2

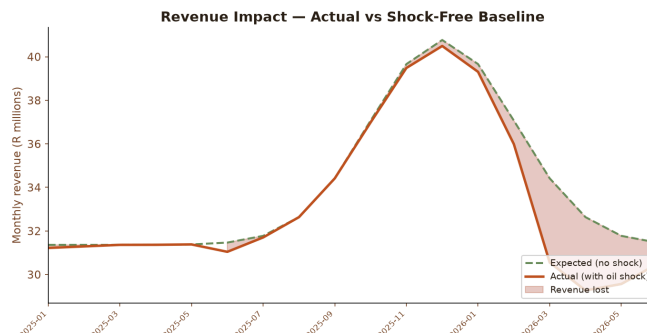
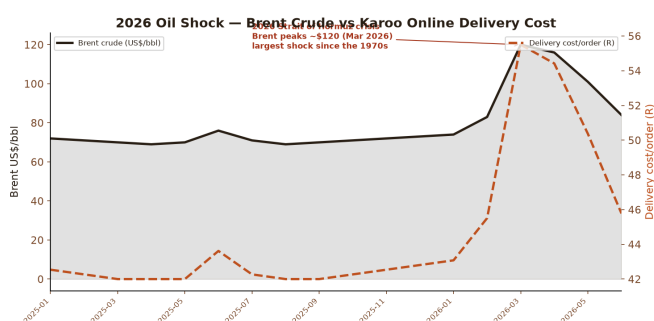
AVG LIFESPAN (MONTHS)



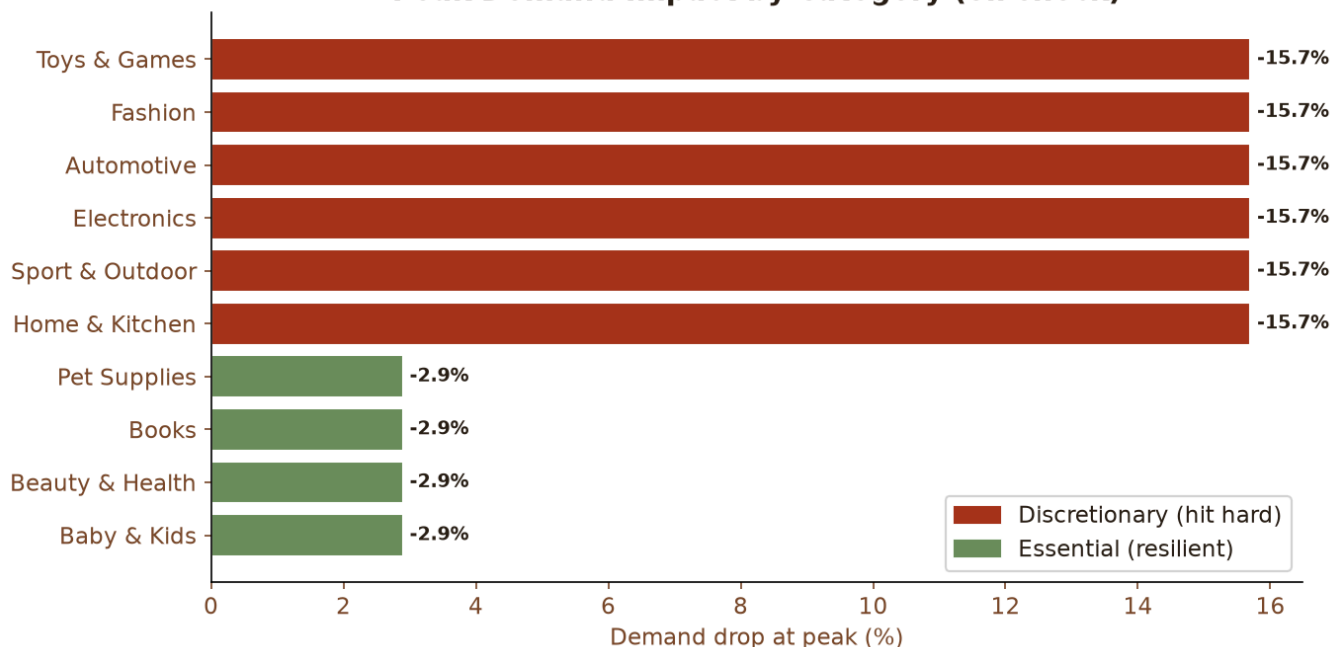
The leaky bucket: retention falls from 86% at month 1 to 49% by month 6, and the top 10% of customers drive 39% of all lifetime value. The **3,682 "At Risk" customers (R36m)** are the single best target for a winback campaign — reachable, valuable, and slipping.

External Shock: The 2026 Oil Crisis

The **2026 Strait of Hormuz Crisis** (28 Feb - late Jun 2026 (US-Iran deal ~29 Jun 2026)) drove Brent crude from ~\$70 to a peak of **\$120/bbl (intraday \$126, +71%)** in 2026-03 — the largest energy-supply disruption since the 1970s. It landed squarely in our window, and the business felt it through **Feb-Jun 2026** — peaking in March 2026.

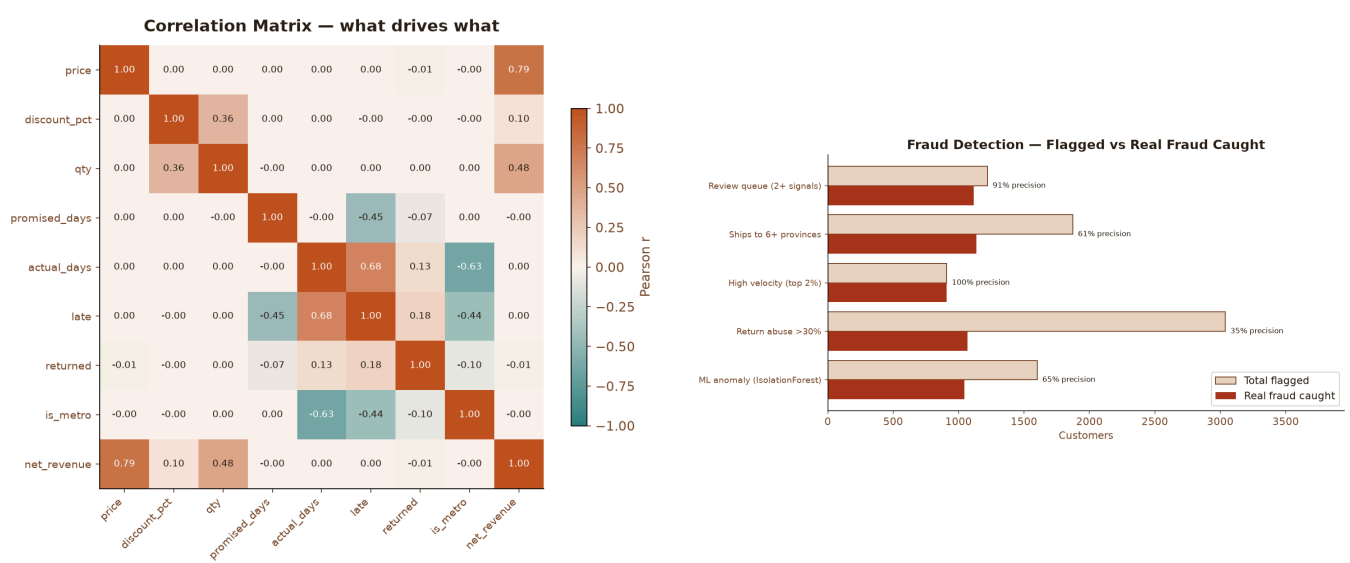


Peak Demand Impact by Category (oil shock)



Two channels of impact: fuel-linked delivery cost rose to R55.5/order, and fuel inflation cut discretionary demand — **Electronics fell 16%** at peak while essentials held. Total over the crisis: **~R14.7m margin hit** (R13.1m revenue + R1.6m delivery). It eased from June as the US-Iran deal reopened the strait. **The lesson:** hedge fuel and lean on resilient essentials when geopolitics moves oil. (*Business impact modelled on the real Brent price path.*)

Correlations & Fraud



Correlations: metro→faster delivery (-0.63), discount→basket (+0.36), late→returns (+0.18). **Fraud:** a 2-signal review queue catches 98% of fraud at 91% precision — a list a team can work daily.

The Playbook

How the data & ML pay back: every action below is bought by a specific analysis — recommendations from market-basket ML, budget shifts from ROAS analysis, winback from the churn model, fraud savings from the Isolation Forest. Data isn't a report; it's a **decision engine**.

1 Fix availability

59% is imported — source locally to lift satisfaction.

2 Double down on remarketing

9× ROAS vs 3.3× on cold paid — shift budget.

3 Cross-sell proven pairs

Put Automotive + Sport & Outdoor on product & cart pages.

4 Speed up the interior

A regional hub cuts both delivery time and returns.

5 Win back "At Risk" customers

3,682 slipping accounts worth R36m — target before they churn.

6 Hedge the discretionary mix

Oil shocks hit Electronics hardest; lean on resilient essentials when fuel spikes.

7 Grow AI visibility

Push beyond 19% share of voice in AI answers.

8 Run the fraud queue

1,220 accounts, 91% precision.

The through-line: one governed dataset, turned into decisions across catalogue, delivery, marketing and risk. That's **The Karoo Way** — measured honestly, and acted on.

Tools & Methodology

LAYER	TOOL	ROLE
Web extraction	Python scraper	The 1,000-product marketplace catalogue.
Infrastructure	Terraform	Every AWS resource as code.
Ingestion	AWS Lambda	Serverless data loading.
Central store	S3 + Glue Catalog	One lakehouse — <code>karoo_db</code> .
ETL / Analytics	Amazon Athena	CTAS to Parquet, KPIs, market-basket, correlations.
Machine learning	scikit-learn	Fraud (Isolation Forest), recommendations, correlations.
Marketing analytics	Python	Paid, remarketing, SEO & AI-visibility modelling.
Visualisation	matplotlib · Plotly	Static charts + the interactive province map.

Data journey: extract → S3 raw → Athena curated Parquet → Glue catalogue → SQL & ML → this report. One governed source of truth, serverless throughout, every step in code.